discussion session: Feb. 26, 2024

textbook: 7.4 Integration of rational functions by partial fractions

7.5 Strategy for integration

ex 1: $\int \frac{x-12}{x^2-4x}\,dx$

(7.4 #8)

partial fractions: $\frac{x-12}{x^2-4x} = \frac{x-12}{x(x-4)} = \frac{A}{x} + \frac{B}{x-4} = \frac{A(x-4)+Bx}{x(x-4)}$

$\Rightarrow A(x-4)+Bx = x-12$

$Ax-4A+Bx = x-12$

$(A+B)x-4A = x-12$

$\Rightarrow \begin{cases} A+B = 1 & \Rightarrow B = 1-A = 1-3 = -2 \\ -4A = -12 & \Rightarrow A = 3 \end{cases}$

$\int \frac{x-12}{x^2-4x}\,dx = \int\left(\frac{3}{x} - \frac{2}{x-4}\right)dx$

$= \boxed{3\ln|x| - 2\ln|x-4| + C}$

ex 2: $\int \frac{3x^2-x+8}{x^3+4x}\,dx$

partial fractions: $\frac{3x^2-x+8}{x^3+4x} = \frac{3x^2-x+8}{x(x^2+4)} = \frac{A}{x} + \frac{Bx+C}{x^2+4} = \frac{A(x^2+4)+x(Bx+C)}{x(x^2+4)}$

$\Rightarrow A(x^2+4)+x(Bx+C) = 3x^2-x+8$

$Ax^2+4A+Bx^2+Cx = 3x^2-x+8$

$(A+B)x^2+Cx+4A = 3x^2-x+8$

$\Rightarrow \begin{cases} A+B = 3 & \Rightarrow B = 3-A = 3-2 = 1 \\ C = -1 & \\ 4A = 8 & \Rightarrow A = 2 \end{cases}$

$\int \frac{3x^2-x^2+8}{x^3+4x}\,dx = \int\left(\frac{2}{x} + \frac{x-1}{x^2+4}\right)dx$

$= \int\left(\frac{2}{x} + \frac{x}{x^2+4} - \frac{1}{x^2+4}\right)dx$

$= 2\ln|x| + \int \frac{x}{x^2+4}\,dx - \frac{1}{2}\tan^{-1}\left(\frac{x}{2}\right)$

$= \boxed{2\ln|x| + \frac{1}{2}\ln|x^2+4| - \frac{1}{2}\tan^{-1}\left(\frac{x}{2}\right) + C}$

or, see that

$\int \frac{1}{x^2+4}\,dx = \frac{1}{4}\int \frac{1}{\left(\frac{x}{2}\right)^2+1}\,dx$

$= \frac{1}{4}\int \frac{2}{u^2+1}\,du = \frac{1}{2}\tan^{-1}u + C$

for u = x/2, dx=dx/2

"$\int \frac{1}{x^2+a^2}\,dx = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C$"

$\left[\begin{array}{l} u = x^2+4 \\ du = 2x\,dx \Leftrightarrow dx = \frac{du}{2x} \end{array}\right]$

so $\int \frac{x}{x^2+4}\,dx = \frac{1}{2}\int \frac{du}{u} = \frac{1}{2}\ln|u|$

$= \frac{1}{2}\ln|x^2+4|$

ex 3: $\int \frac{x^3-4x+1}{x^2-3x+2}\,dx$

polynomial division:

$$\begin{array}{r} x+3 \\ x^2-3x+2 \overline{\big) x^3-4x+1} \\ -(x^3-3x^2+2x) \\ \hline 3x^2-6x+1 \\ -(3x^2-9x+6) \\ \hline 3x-5 \end{array} \quad + \frac{3x-5}{x^2-3x+2}$$

$$\int \frac{x^3-4x+1}{x^2-3x+2}\,dx = \int \left(x+3+\frac{3x-5}{x^2-3x+2}\right) dx$$

$$= \int (x+3)\,dx + \int \frac{3x-5}{x^2-3x+2}\,dx$$

$$= \int (x+3)\,dx + \int \left(\frac{1}{x-2}+\frac{2}{x-1}\right) dx$$

$$= \boxed{\frac{x^2}{2}+3x+\ln|x-2|+2\ln|x-1|+C}$$

partial fractions:

$$\frac{3x-5}{x^2-3x+2} = \frac{3x-5}{(x-2)(x-1)} = \frac{A}{x-2}+\frac{B}{x-1} = \frac{A(x-1)+B(x-2)}{(x-2)(x-1)}$$

$$\Rightarrow A(x-1)+B(x-2) = 3x-5$$

$$Ax-A+Bx-2B = 3x-5$$

$$(A+B)x-A-2B = 3x-5$$

$$\Rightarrow \begin{cases} A+B=3 \\ -A-2B=-5 \end{cases} \Rightarrow A = 3-B = 3-2 = 1$$

$$+ \quad -B=-2 \Rightarrow B=2$$

ex 4: $\int \frac{3\,dx}{(x^2-1)^2}$

partial fractions: $\frac{3}{(x^2-1)^2} = \frac{3}{(x-1)^2(x+1)^2} = \frac{A}{x-1}+\frac{B}{(x-1)^2}+\frac{C}{x+1}+\frac{D}{(x+1)^2} = \frac{A(x-1)(x+1)^2+B(x+1)^2+C(x-1)^2(x+1)+D(x-1)^2}{(x-1)^2(x+1)^2}$

$$\Rightarrow A(x-1)(x+1)^2+B(x+1)^2+C(x-1)^2(x+1)+D(x-1)^2 = 3$$

$$A(x^2-1)(x+1)+B(x^2+2x+1)+C(x^2-1)(x-1)+D(x^2-2x+1) = 3$$

$$A(x^3+x^2-x-1)+B(x^2+2x+1)+C(x^3-x^2-x+1)+D(x^2-2x+1) = 3$$

$$(A+C)x^3+(A+B-C+D)x^2+(-A+2B-C-2D)x-A+B+C+D = 3$$

$$\Rightarrow \begin{cases} A+C=0 & \Rightarrow A=-C \\ A+B-C+D=0 \\ -A+2B-C-2D=0 & \Rightarrow -A+2B+A-2D = 2B-2D=0 \Rightarrow B=D \\ -A+B+C+D=3 \end{cases}$$

Given $A=-C$ and $B=D$, the 2nd + 4th equations become

$$\begin{cases} A+B+A+B = 2A+2B = 0 \Rightarrow A=-B \\ -A+B-A+B = -2A+2B = 3 \Rightarrow 2B+2B = 4B = 3 \Rightarrow B = \frac{3}{4} \end{cases}$$

so, $A=-B=-\frac{3}{4}$, $C=-A=\frac{3}{4}$, and $D=B=\frac{3}{4}$.

$$\int \frac{3\,dx}{(x^2-1)^2} = \int \left(\frac{-3/4}{x-1}+\frac{3/4}{(x-1)^2}+\frac{3/4}{x+1}+\frac{3/4}{(x+1)^2}\right) dx$$

$$= \boxed{-\frac{3}{4}\ln|x-1|-\frac{3}{4}\cdot\frac{1}{x-1}+\frac{3}{4}\ln|x+1|-\frac{3}{4}\cdot\frac{1}{x+1}+C}$$

Note: $\begin{bmatrix} u=x+1 \\ du=dx \end{bmatrix}$ so $\int \frac{3/4}{u^2}\,du = -\frac{3}{4}\cdot\frac{1}{u}$

or $\begin{bmatrix} u=x-1 \\ du=dx \end{bmatrix}$